

Homework (Jalapeno)

- Q1.** The ancient Egyptian Rhind Papyrus contains a problem that can be modeled by $x^2 + 5x + 6 = 0$. What is the discriminant of this quadratic equation?
- (A) 1
(B) 13
(C) 5
(D) -1
(E) 25
- Q2.** In the quadratic equation $x^2 - 4x + 4 = 0$, what is the number and type of solutions?
- (A) Two distinct real solutions
(B) One repeated real solution
(C) No real solutions
(D) Two complex solutions
(E) One real and one complex solution
- Q3.** The Great Pyramid of Giza's height can be modeled by $h(t) = -t^2 + 10t + 20$, where h is the height and t is time. What is the discriminant of this quadratic function?
- (A) 60
(B) 100
(C) 80
(D) -20
(E) 20
- Q4.** Solve for x in the equation $x^2 + 2x - 6 = 0$ using the quadratic formula. What are the solutions?
- (A) $x = -1 \pm \sqrt{7}$
(B) $x = 1 \pm \sqrt{7}$
(C) $x = -3 \pm \sqrt{2}$
(D) $x = 2 \pm \sqrt{3}$
(E) $x = -1 \pm \sqrt{3}$
- Q5.** The ancient Greeks used the quadratic formula to solve problems related to the construction of the Parthenon. If a quadratic equation has a discriminant of -3 , what can be concluded about its solutions?
- (A) Two distinct real solutions
(B) One repeated real solution
(C) No real solutions
(D) Two complex solutions
(E) One real and one complex solution
- Q6.** The quadratic equation $x^2 - 2x - 3 = 0$ has two solutions. What is the product of these solutions?
- (A) -3
(B) 2
(C) -2
(D) 3
(E) 1
- Q7.** An ancient Babylonian tablet contains a problem that can be modeled by $x^2 + 2x + 1 = 0$. What is the solution to this equation?
- (A) $x = -1$
(B) $x = 1$
(C) $x = -1 \pm \sqrt{2}$
(D) $x = 2 \pm \sqrt{3}$
(E) $x = -2 \pm \sqrt{2}$
- Q8.** The quadratic equation $x^2 + 4x + 4 = 0$ has one solution. What is the value of this solution?
- (A) $x = -2$
(B) $x = 2$
(C) $x = -1$
(D) $x = 1$
(E) $x = 0$

Open Ended Questions (FRQ)

- Q9.** The ancient Egyptians built a pyramid with a height that can be modeled by $h(t) = -t^2 + 12t + 20$, where h is the height and t is time. Use the quadratic formula to find the time when the height is 0. Show your work and explain your answer.
- Q10.** A ancient Greek astronomer used the quadratic formula to calculate the time it takes for a planet to orbit the sun. If the quadratic equation is $x^2 - 6x + 8 = 0$, use the quadratic formula to find the solutions and explain what they represent in the context of the problem.

Chef's Tips

- Tip 1: Make sure students show their work and explain their answers.
- Tip 2: Encourage students to check their solutions by plugging them back into the original equation.
- Tip 3: Remind students to use the quadratic formula correctly and to simplify their answers.